👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3C

Convert Measurement Units

Lesson 3-10 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can convert measurement units using ratios and conversion factors.

Language Objective: I can explain my conversion using the words conversion factor, unit, equivalent, and ratio.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Conversion Factor · **Unit** · **Equivalent** · **Ratio** · **Customary units** · **Metric units** · **Ratio table**

Level 3: try the blanks from memory first, then check the bank.

1 A ratio equal to 1 that relates two units, like 12 inches / 1 foot, is a _____.

2. A standard amount used to measure, like a foot or a gram, is a _____.

3. Two measurements that name the same amount, like 1 foot and 12 inches, are _____.

4. A comparison of two quantities is a _____.

5. Units like inches, feet, and pounds used mainly in the U.S. are _____.

6. Units like meters, grams, and liters based on tens are _____.

7. A table that lists equivalent ratios in rows or columns is a _____.

Write About the Math

Will the cabinet fit through the doorway? Show your conversion to prove it?
¿Cabrá el mueble por la puerta? Muestra tu conversión para probarlo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Conversion Factor** — *Factor de conversión*
- ☐ **Unit** — *Unidad*
- ☐ **Equivalent** — *Equivalente*
- ☐ **Ratio** — *Razón*
- ☐ **Customary units** — *Unidades usuales (sistema inglés)*

Model: *A foot is bigger than an inch, so I must convert to compare or combine the measurements.* — Students explain a foot is the larger customary unit (12 inches) and that mixed units must be converted to the same unit before working with them.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **7 feet = 7 × ___ = ___ inches. Since ___ inches is ___ than 80 inches, the cabinet ___ fit through the doorway.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Conversion Factor
2. **Notes 2:** Unit
3. **Notes 3:** Equivalent
4. **Notes 4:** Ratio
5. **Notes 5:** Customary units
6. **Notes 6:** Metric units
7. **Notes 7:** Ratio table

8. **Watch for this mistake:** *A common mistake in Convert Measurement Units is applying the conversion factor in the wrong direction — for example, converting 4 feet to inches by dividing (4 ÷ 12 = 0.33 inches) instead of multiplying (4 × 12 = 48 inches), or converting 80 ounces to pounds by multiplying (80 × 16 = 1,280 pounds) instead of dividing (80 ÷ 16 = 5 pounds). Before you answer, ask: am I changing to a SMALLER unit (there will be MORE of them, so multiply) or a BIGGER unit (there will be FEWER of them, so divide)?*
9. **Try It 1:** D. 80 ounces — *Ounces are smaller than pounds, so multiply: 5 × 16 = 80 ounces.*
10. **Try It 2:** A. 2 gallons — *Gallons are bigger than quarts, so divide: 8 ÷ 4 = 2 gallons.*